

Muhammad Umar Farooq, Ph.D.,

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Research Areas: Machine Learning (ML), Artificial Intelligence (AI), Automatic Speech Recognition (ASR), self-supervised and multilingual learning, dialect identification

Professional Experience

- **Applied Research Scientist** July 2024 - Present
Emotech Ltd. United Kingdom
 - Develop production-scale speech-to-text (STT) systems for real-world deployment, balancing accuracy, latency, and cost trade-offs.
 - Collaborate with cross-functional engineering teams to deploy and scale ML models through production MLOps pipelines for high-throughput speech processing.
 - Reduced model size by 3x and inference latency by 1.5–2x through model compression and architecture optimization, maintaining recognition quality on domain-adapted STT systems.
 - Architected a high-throughput streaming inference pipeline with dynamic batching, reducing inference infrastructure costs by 80%.
 - Leveraged large language models (LLMs) for synthetic data generation, text augmentation, and retrieval-based enhancement of ASR performance.
- **Doctoral researcher** Feb 2021 - July 2024
LivePerson Centre for SLT, Dept. of Computer Science University of Sheffield, UK.
 - Investigated cross-lingual representation sharing and proposed multilingual model fusion techniques for improved generalisation across languages.
 - Developed data augmentation and knowledge distillation approaches to bridge performance gaps in low-resource and cross-lingual ASR settings.
 - Designed progressive unsupervised domain adaptation methods for self-supervised ASR, improving robustness across acoustic and domain shifts.
 - Trained large-scale neural speech models using distributed data-parallel (DDP) training across multi-GPU, multi-node GPU clusters in an HPC (SLURM-managed) environment.
- **Pre-doc Intern** Sep 2020 - Dec 2020
Institute of Formal and Applied Linguistics (UFAL) Charles University, Prague.
 - Funded by **European Union (EU)**
 - Adapted language models for ASR in the **EU Horizon 2020** project European Live Translator ([ELITR](#))
- **Research Assistant** Jun 2017 - Aug 2020
Center for Language Engineering (CLE) University of Engineering and Technology Lahore
 - Designed a corpus and developed semi-supervised data collection tools for Urdu speech.
 - Built the first Urdu LVCSR system and [deployed it online](#) with a public [API](#).
 - Adapted Urdu ASR for Urdu–English code-switching and for Punjabi via zero-shot transfer.
 - Developed a real-time Urdu–English speech-to-speech (S2S) translation system.

Education

- **Ph.D. Computer Science** Feb 2021 - May 2024
University of Sheffield Sheffield, UK.
 - **Supervisor:** Prof. Thomas Hain
 - **Industry partner:** LivePerson Inc.
 - **Research areas:** speech and language processing, foundational models, multilingualism
 - **Thesis title:** Non-linear cross-lingual mappings for low-resource speech recognition

- M.Sc. Electrical Engineering**
Sep 2017 - Jun 2019
University of Engineering & Technology
Lahore, Pakistan.
 - **CGPA:** 3.68/4.00
 - **Research areas:** speech processing, speaker diarisation
- B.Sc. Electrical Engineering**
Oct 2013 - May 2017
University of Engineering & Technology
Lahore, Pakistan.
 - **CGPA:** 3.42/4.00

Selected Publications

For full list, please visit the [Google Scholar profile](#) or my [publications web page](#)

- [1] CTC-DID: CTC-Based Arabic dialect identification for streaming applications**
M. U. Farooq and O. Saz
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026
- [2] Enhancing Low-resource Speech Recognition with Non-linear Cross-lingual Mappings**
M. U. Farooq and T. Hain
IEEE Transactions on Audio, Speech and Language Processing (TASLP), 2025
- [3] MUST: A Multilingual Student-Teacher Learning Approach for Low-Resource Speech Recognition**
M. U. Farooq, R. Ahmad and T. Hain
IEEE Automatic Speech Recognition and Understanding (ASRU), 2023
- [4] Learning Cross-lingual Mappings for Data Augmentation to Improve Low-Resource Speech Recognition**
M. U. Farooq and T. Hain
Interspeech, 2023
- [5] Non-Linear Pairwise Language Mappings for Low-Resource Multilingual Acoustic Model Fusion**
M. U. Farooq, D. A. H. Narayana and T. Hain
Interspeech, 2022
- [6] Investigating the Impact of Cross-lingual Acoustic-Phonetic Similarities on Multilingual Speech Recognition**
M. U. Farooq and T. Hain
Interspeech, 2022
- [7] Improving Large Vocabulary Urdu Speech Recognition System using Deep Neural Networks**
M. U. Farooq, F. Adeeba, S. Rauf and S. Hussain
Interspeech, 2019

Teaching and Mentoring Experience

- Teaching Assistant**
Feb 2024 - June 2024
University of Sheffield, UK.
- Graduate Teaching Assistant (GTA)**
Sep 2022 - Feb 2024
University of Sheffield, UK.
- Mentoring / Supervising Interns**
LivePerson Centre for SLT, University of Sheffield

Skillset

Technical Skills

- **Languages:** Python, Shell, Rust, Java, C, MATLAB, L^AT_EX
- **Frameworks:** PyTorch, Hugging Face Transformers, NVIDIA NeMo, SpeechBrain, Fairseq
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib
- **ML Systems:** Distributed training (PyTorch DDP), multi-GPU training, model compression, inference optimization, streaming inference, MLOps
- **Developer Tools:** Git, Docker, OpenAI Codex, Gemini CLI

Language Proficiency

English (professional proficiency), Urdu (native), Punjabi (native), Hindi (spoken proficiency), Arabic (beginner)

Achievements, Awards and Honours

- **Postgraduate Publication Scholarship, University of Sheffield** Mar 2024 - May 2024
- Submitted my **PhD in 3 years**
- Published **6 research papers** during 3 years of my PhD (4 as a first author and 2 collaborated works)
- **Departmental Scholarship Award, University of Sheffield** (*Fully funded Ph.D.*) Feb 2021 - Jan 2024